Data Sharing Questionnaire

This questionnaire helps you find which management approach fits your data ecosystem project, and describes the project in a common vocabulary. It is meant for the enterprise architect involved in the project, alone or with a few colleagues. No prior knowledge of data ecosystems or of the management approaches is needed.

The questionnaire has 23 questions in five sections:

1. **Data:** what the data is, where it comes from, how sensitive it is and how often it changes.
2. **Organization:** why the data is shared, with whom, and what is given in return.
3. **The organization(s) taking part:** what they aim for, how they are led, how far their processes are harmonized and how regulated their markets are. This section has two versions; your answer to Q7 tells you which one to answer.
4. **Network:** how the data is exchanged in practice.
5. **Governance:** who decides about the shared data once it is shared.

Before you start. Answer for the project as it has been agreed today: what is already running, and what has been agreed but not yet built. Do not answer for ideas that have not been agreed; choose "I am not sure / not decided yet" instead. Every question ends with that option, so you never have to guess.

○ choose one answer □ choose all that apply

Section 1. Data

This section is about the data itself. What the data is, what state it is in when it leaves your side, where it came from, how sensitive it is and how often it has to be refreshed all put limits on how it can be shared and on what has to be arranged around it. Answer for the data in this specific project rather than for your organization as a whole.

Q1. Which of the following categories describe the data the project shares, or will share?
(choose all that apply)

- □ **Process data:** data from organizational processes, research or development (for example, production line logs, maintenance records, R&D test results).
- □ **Product data:** data from smart products or sensors, sometimes called Big Data (for example, vibration readings from a pump, GPS traces from a truck fleet, machine telemetry).
- □ **Environment data:** data about the surroundings (for example, weather forecasts, air quality readings, soil or terrain maps).
- □ **Customer data:** data specifically about customers (for example, order history, contact details, subscription or usage behaviour).
- □ **Metadata:** data that describes another dataset (for example, a data catalogue entry, column definitions, the owner and timestamp of a file).
- □ **Other:** _____
- □ **I am not sure / not decided yet**

Q2. In which state is the data at the moment it is shared? (choose all that apply)

- □ **Raw data:** exactly as received from the source (for example, unfiltered sensor readings, untouched machine log files).
- □ **Modified data:** data that has been preprocessed (for example, duplicates removed, missing values filled in, units standardized, labels added).
- □ **Enriched data:** the original data plus extra information from outside the original source (for example, sales figures combined with regional demographics, sensor readings joined with weather data).
- □ **Processed data:** data ready to be analysed by algorithms or ML models (for example, a cleaned feature table, a pre-aggregated dataset).
- □ **Other:** _____
- □ **I am not sure / not decided yet**

Q3. From which sources does the shared data originate? (choose all that apply)

- □ **Generated data:** generated by our own operations, sensors and equipment (for example, our own production line, our own machines and vehicles).
- □ **Acquired data:** obtained from data generated by other organizations (for example, bought from a data provider, received from a supplier under an agreement).
- □ **Customer provided data:** collected directly from our own customers (for example, forms customers fill in, survey answers, app usage).
- □ **Freely available data:** data that is openly published and can be used without an agreement (for example, open government datasets, public weather or map data).
- □ **I am not sure / not decided yet**

Q4. Which level of privacy does the data require? (choose one)

- ○ **Public data:** no restrictions apply, and the data may be shared without restriction (for example, published open datasets, public timetables, published environmental readings).
- ○ **Internal data:** the data can be shared within the domain, but requires restrictions when shared outside the domain (for example, internal performance reports, team dashboards).
- ○ **Confidential data:** as internal data, but more strongly restricted: only authorized personnel may access it (for example, contract terms, pricing, unreleased product designs).
- ○ **Sensitive data:** the data contains personally identifiable information, so sharing it requires legal compliance (anonymization, pseudonymization) (for example, employee records, patient data, customer personal data under GDPR).
- ○ **I am not sure / not decided yet**

Q5. How frequently does the data have to be updated once it is shared? (choose all that apply)

- □ **Static:** no updates are required after sharing (for example, a one-off historical export, a fixed reference table).
- □ **Up to date:** repeated updates on a schedule (for example, a daily or weekly batch upload).
- □ **(near) Real time:** a live stream or continuous feed (for example, sensor telemetry pushed the moment it is measured).
- □ **I am not sure / not decided yet**

Section 2. Organization

This section is about why the data is shared and with whom: what the organization seeks from the exchange, which parties take part in it, and what is given or received in return. Answer for this specific project rather than for your organization as a whole.

Q6. What motivates the sharing of this data? (choose all that apply)

- □ **Economic:** money or efficiency (for example, new revenue from selling data, cost savings, better forecasting).
- □ **Social and environmental:** to contribute to social or environmental goals (for example, sustainability reporting, emissions tracking, corporate social responsibility programmes).
- □ **Legal:** legislation or a regulator requires the data to be shared (for example, GDPR obligations, EU Data Act, regulator or safety reporting).
- □ **Cultural:** to improve the culture around data sharing within the organization (for example, breaking down silos between departments, building trust in shared data).
- □ **Other:** _____
- □ **I am not sure / not decided yet**

Q7. With whom is the data shared, or to be shared? (choose one)

- ○ **Intra-organizational:** the data stays inside our own organization (for example, between our production, sales and logistics departments).
- ○ **Same industry:** only with organizations in the same industry (for example, a supplier sharing operational data with a manufacturer).
- ○ **Cross-industry:** across sector boundaries (for example, sharing meteorology data with agriculture or aerospace companies).
- ○ **I am not sure / not decided yet**

Q8. What does your organization expect in return for sharing the data? (choose all that apply)

- □ **Financial reward:** a monetary return (for example, a licence fee, a price per dataset).
- □ **Virtual assets:** digital assets that cannot be directly converted into cash (for example, platform credits, tokens, points).
- □ **Data in return:** data is received in exchange for the data shared (for example, we share production data, the partner shares market data).
- □ **Services:** additional or new services from the receiving party (for example, a free analytics dashboard, a maintenance or benchmarking service).
- □ **(Increased) Reputation:** greater brand awareness, trust or credibility (for example, recognized as a trusted partner, publicity as a frontrunner in the sector).
- □ **Nothing in return:** no return is expected (for example, the data is shared because a regulator requires it, or it is published openly).
- □ **Other:** _____
- □ **I am not sure / not decided yet**

Q9. Which form of compensation is expected for the data? (choose one)

- ○ **Free:** the data is shared free of charge (for example, internal sharing between departments, open data, legally mandated reporting).
- ○ **Fixed:** a single, fixed price, paid at once or in instalments (for example, EUR 5,000 for a dataset).
- ○ **Subscription based:** a recurring price every set period (for example, a monthly or yearly access fee).
- ○ **Usage based:** the compensation varies with how much the data is used (for example, a price per API call, per record downloaded).
- ○ **Revenue based:** compensation based on the revenue generated from the shared data (for example, 8% of the revenue generated from the data, or royalties).
- ○ **Hybrid:** a combination of the models mentioned above (for example, a base subscription plus a charge per API call).
- ○ **I am not sure / not decided yet**

Section 3a. The Organization

Answer this version if you answered Q7 "Intra-organizational".

This section is about the organization in which the project runs: what it emphasizes, how centralized it is, how far its processes are harmonized, and how regulated its market is. Answer for the organization, or for the part of it that the project spans.

Q10. Which performance objective does the organization emphasize most? (choose one)
- **Profit:** earning more from the current business, mainly by keeping costs down (for example, margin targets, cost-reduction programmes).
- **Asset utilization:** getting more out of what the organization already owns (for example, using plants, vehicles or systems more fully).
- **Growth:** expanding into new products, markets or customers (for example, revenue growth targets, new business lines).
- **I am not sure / not decided yet**

Q11. How centralized is the organization, or the part of it that the project spans? (choose one)
- **Centralized:** most decisions are taken at the top (for example, headquarters sets budgets, priorities and policies for all units).
- **Decentralized:** the business units or regions decide largely for themselves (for example, each division runs its own budget and priorities).
- **I am not sure / not decided yet**

Q12. How harmonized are the business processes that the shared data supports, across the parts of the organization? (choose one)
- **Globally harmonized:** the units follow the same processes (for example, one order-to-cash process for all countries).
- **Local processes:** each unit follows its own processes (for example, country-specific sales or registration procedures).
- **I am not sure / not decided yet**

Q13. How regulated is the market in which the organization operates? (choose one)
- **Highly regulated:** the products, processes or data are subject to extensive sector regulation and reporting (for example, pharmaceuticals, chemicals, banking, energy).
- **Little or no regulation:** little or no sector-specific regulation applies beyond general law such as data protection (for example, consulting, software development, non-food retail).
- **I am not sure / not decided yet**

Section 3b. The Participating Organizations

Answer this version if you answered Q7 "Same industry" or "Cross-industry". If you answered Q7 "I am not sure / not decided yet", skip Section 3 and go to Section 4.

This section is about the participating organizations taken together: what the collaboration aims at, how the collaboration is led, how far the participants' processes are harmonized, and how regulated their markets are.

Q10. What do the participating organizations mainly aim for with the collaboration? (choose one)
- **Profit:** lowering the cost of what the participants already do (for example, fewer manual document exchanges along a supply chain).
- **Asset utilization:** using capacity the participants already have more fully (for example, shared berths, trucks or warehouse space).
- **Growth:** new services or markets the participants could not reach alone (for example, a new service built on the combined data).
- **I am not sure / not decided yet**

Q11. Apart from the shared data, how is the business collaboration between the participating organizations led? (choose one)
- **Centralized:** one organization leads the collaboration (for example, a manufacturer and its suppliers, a port authority and the companies in its port).
- **Decentralized:** no organization leads, and the participants take part on an equal footing (for example, an association of companies of similar standing).
- **I am not sure / not decided yet**

Q12. How harmonized are the business processes that the shared data supports, across the participating organizations? (choose one)

- ○ **Globally harmonized:** the participants follow the same processes (for example, an agreed industry procedure for booking, ordering or customs clearance).
- ○ **Local processes:** each organization follows its own processes, and only the exchange between them is aligned (for example, each partner plans its own operations and sends the agreed messages).
- ○ **I am not sure / not decided yet**

Q13. How regulated are the markets in which the participating organizations operate? If they differ, answer for the most regulated one. (choose one)

- ○ **Highly regulated:** the products, processes or data are subject to extensive sector regulation and reporting (for example, pharmaceuticals, chemicals, banking, energy).
- ○ **Little or no regulation:** little or no sector-specific regulation applies beyond general law such as data protection (for example, consulting, software development, non-food retail).
- ○ **I am not sure / not decided yet**

Section 4. Network

This section is about how the exchange works in practice: the route the data takes between the parties, where the data sits once it is shared, what has to line up on both sides for it to be understood, how access is regulated, what sets an exchange off, and how many parties are involved.

Q14. By which route is the data exchanged between the participants? (choose one)

- ○ **Peer to peer:** the data passes directly from one party to the other, with no intermediary (for example, our system calls theirs directly, or they call an API we run ourselves).
- ○ **Platform based:** the exchange happens through the intermediation of a platform (for example, a shared portal, or a broker that sits between provider and consumer).
- ○ **Other:** _____
- ○ **I am not sure / not decided yet**

Q15. Where does the data reside when it is shared? (choose one)

- ○ **Centralized:** copied into one shared location, from which consumers retrieve it (for example, all departments load their data into one shared database that others query).
- ○ **Distributed:** the data stays where it is produced and is accessed at the source through an agreed interface (for example, each domain or organization exposes its own data through a connector or API; nothing is duplicated centrally).
- ○ **Hybrid:** some data is copied into a central store, other data stays at the source and is accessed there.
- ○ **I am not sure / not decided yet**

Q16. Which levels of interoperability does the exchange require between the parties? (choose all that apply)

- □ **Legal interoperability:** compatible legal frameworks, standards and regulations (for example, matching contract clauses, GDPR-compliant terms, agreed usage policies).
- □ **Organizational interoperability:** a shared understanding of processes, objectives and resources (for example, agreed roles, responsibilities and escalation paths between the parties).
- □ **Semantic interoperability:** consistency and interpretability across diverse sources and systems (for example, a shared vocabulary or ontology, so that “customer” or “downtime” means the same thing on both sides).
- □ **Technical interoperability:** the same technical data exchange across systems (for example, matching formats and protocols, JSON over REST, OPC UA, agreed schemas).
- □ **I am not sure / not decided yet**

Q17. On what basis is access to the data granted? (choose one)

- **Agreement based:** access terms are defined between you and the other party (for example, a bilateral data sharing contract, an NDA with data clauses).
- **Application based:** access is regulated through those who use the specific application (for example, only users logged into the shared dashboard can see the data).
- **Trust based:** access is granted on the basis of trust (for example, known partners are granted access without a formal contract or technical gate).
- **Open:** no access restrictions apply, and anyone may access the data (for example, published as open data).
- **I am not sure / not decided yet**

Q18. What triggers an exchange of data? (choose one)

- **On demand:** sharing is triggered when the consumer asks for the data (for example, the consumer queries the API when they need it).
- **Event based:** triggered by an event specified by the provider (for example, end of day, the dataset reaching a certain size, an alarm or threshold being crossed).
- **Continuous:** sharing data in real time or at regular intervals (for example, a live stream, or an automatic push every 15 minutes).
- **Other:** _____
- **I am not sure / not decided yet**

Q19. Which pattern do the data sharing relationships follow? (choose one)

- **One-to-one:** one party shares data with one other party (for example, one supplier sends data to one manufacturer).
- **One-to-many:** one party shares data with many other parties (for example, one provider publishes a dataset that many consumers use).
- **Many-to-one:** many parties share data with one other party (for example, many suppliers report into one OEM).
- **Many-to-many:** many parties share data with many other parties (for example, an ecosystem where every participant both provides and consumes).
- **I am not sure / not decided yet**

Q20. How dependent are the parties on one another's data? (choose one)

- **Loosely coupled:** each party can continue to operate if the data stops arriving (for example, the shared data improves a forecast, but the process still runs without it).
- **Tightly coupled:** the parties depend on each other's data to operate (for example, a production schedule that cannot be made without the partner's live data).
- **I am not sure / not decided yet**

Section 5. Governance

This section is about control over the data once it has been shared: who decides what may be done with it afterwards, who holds the authority to set the rules for it and see that they are followed, and how decisions about it are reached. These questions are often still open early in a project; please use the "I am not sure / not decided yet" option rather than guessing.

Q21. Once the data has been shared, can the party that provided it still limit what is done with it? (choose one)

- **Yes:** the provider sets conditions on how the data may be used (for example, usage policies travel with the data, or the receiver may only use it for an agreed purpose).
- **No, the receiver decides:** whoever receives the data decides how it is used (for example, the departments hand their data over to a central team, or the participants hand theirs to one company, which then decides what is done with it).
- **No conditions at all:** anyone who receives the data may use it freely (for example, the data is published as open data, or made available to every department without conditions).
- **I am not sure / not decided yet**

Q22. When a party wants access to the shared data, who decides whether it gets access? (choose one)

- **Each party, for its own data:** each party decides who may access the data it provides, and any common rules are agreed by the parties together (for example, each domain or organization decides about the data it publishes, and the common standards are agreed in an association in which the parties decide).
- **One party, for all the data:** one party decides for everyone, including about data it did not provide (for example, a central data office or the company that runs the exchange decides who may access which data, or a data trustee that holds the data of several providers).
- **Each party, within common rules set by someone else:** a body that is not one of the parties sets the common rules, and each party decides within them who may access its own data (for example, headquarters sets the data definitions and quality rules for all units, and each unit decides who may use its data).
- **I am not sure / not decided yet**

Q23. How are decisions about the shared data reached? (choose one)

- **Hierarchical:** whoever holds the authority decides on their own and passes the decision down (for example, a manager or a board decides and the teams carry it out).
- **Cooperative:** whoever holds the authority decides together with the others involved (for example, decisions are taken in working groups or committees in which the parties take part).
- **I am not sure / not decided yet**